

Lecture Notes:
Math & Statistics for Data Science
Unit 3 – Calculus

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1 Lecture 1: Approximation, Error, and Derivatives

Learning Goals

- Understand the role of approximation in mathematics and computing.
- Distinguish absolute and relative error.
- Introduce the derivative as a limit of difference quotients.

Content

- **Approximation strategy** (Heath reference): replace complicated problems with simpler ones.
 - Infinite $\rightarrow$ finite, continuous $\rightarrow$ discrete, nonlinear $\rightarrow$ linear.
- **Example:** Earth surface area using $A = 4\pi r^2$.
 - Sphere approximation, empirical radius, truncated π , finite precision.
- **Error types:** absolute vs. relative error.
- **Sources of error:** measurement, model simplifications, floating-point limitations.

- **Error propagation:** chained approximations accumulate error.
- **Derivatives:**
 - Rate of change = slope of secant line.
 - Limit as $\Delta t \rightarrow 0$ gives instantaneous rate:

$$f'(t) = \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}.$$

- Continuous vs. discontinuous functions.

Exercises

- Approximate Earth's surface area with given radius.
- Identify sources of error in real-world models.
- Compute secant slopes for simple functions and compare with derivatives.

2 Lecture 2: Continuity, Differentiability, and Integrals

Learning Goals

- Recognize the difference between continuity and differentiability.
- Apply second derivatives for optimization.
- Understand integrals as both antiderivatives and areas.

Content

- **Continuity vs. discreteness:** discrete data is not truly differentiable.
- **Differentiability:** smoothness, no corners/cusps.
- **Second derivative:** curvature, concavity, optimization.

- $f'(x) = 0$ indicates candidate extrema.
- $f''(x) > 0 \Rightarrow$ local minimum; $f''(x) < 0 \Rightarrow$ local maximum.

- **Integrals:**

- Antiderivative: $F' = f$.
- Area interpretation: Riemann sums (left, right, midpoint, trapezoid).
- Fundamental Theorem of Calculus:

$$\int_a^b f'(x) dx = f(b) - f(a).$$

Exercises

- Differentiate $f(x) = x^2$ and test concavity.
- Approximate area under a curve with Riemann sums.
- Apply second derivative test for maxima/minima.

3 Lecture 3: Multivariable Calculus

Learning Goals

- Extend differentiation and integration to functions of two variables.
- Interpret partial derivatives and multiple integrals.

Content

- **Functions of two variables:** graphs and surfaces.
- **Partial derivatives:**
 - Differentiate with respect to one variable while holding the other constant.
 - Notation: $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$.
- **Multiple integrals:** generalization of Riemann sums to area/volume.

Exercises

- Compute partial derivatives of given functions.
- Evaluate simple double integrals.
- Plot surfaces in Python and interpret slopes of cross-sections.

4 Lecture 4: Chain Rule, Gradient, and Optimization

Learning Goals

- Understand the chain rule and its applications.
- Define the gradient and its role in optimization.
- Connect calculus to machine learning optimization methods.

Content

- **Chain rule:** derivative of composite functions.
 - Application: sequential computations, deep learning backpropagation.
- **Gradient:** vector of all partial derivatives; points in direction of steepest increase.
- **Optimization:**
 - 1D tests: first and second derivative.
 - Gradient descent: iterative minimization.
 - Simulated annealing: stochastic alternative.

Exercises

- Apply the chain rule to composite functions.
- Compute gradients of multivariable functions.
- Implement gradient descent in Python.

Pedagogical Notes

- Emphasize conceptual motivation: approximation, error, rate of change, area.
- Use graphical intuition: secant vs. tangent lines, concavity.
- Incorporate Python labs: numerical derivatives, Riemann sums, gradient descent.
- Connect to data science: error propagation, optimization, and regression as a first ML model.